Keisuke Kamahori

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Education

University of Washington — Seattle, WA
Ph.D. in Computer Science & Engineering — *Jun 2023 – May 2028 (Expected)*
- Advisor: Prof. Baris Kasikci.

The University of Tokyo — Tokyo, Japan
B.Sc. in Information Science — *Apr 2019 – Mar 2023*
- Advisor: Prof. Shinya Takamaeda-Yamazaki.

Publications

Google Scholar: `https://scholar.google.com/citations?user=IMZC2hYAAAAJ`
Semantic Scholar: `https://www.semanticscholar.org/author/Keisuke-Kamahori/2283847574`
(* denotes equal contribution)

Preprints

[P1] **VibeServe: Can AI Agents Build Bespoke LLM Serving Systems?**
Keisuke Kamahori*, Shihang Li*, Simon Peter, Baris Kasikci.
Preprint. (*Presented at DL4C Workshop @ ICML 2026, July 2026.*)

[P2] **SpeedrunBench: Challenging LLM Agents with Video Game Speedrunning.**
Yoshinari Fujinuma, **Keisuke Kamahori**, Ryuto Koike, Abdelrahman Madkour, Varun Prashant Gangal, Monty Bichouna, Martyna Markiewicz, Shivani Jain, Duncan Curtis, Rebecca Qian, Anand Kannappan.
Preprint.

[P3] **ServingStudio: An Integrated Workbench for Simulating, Analyzing, and Optimizing LLM Serving Systems.**
Kan Zhu, Michael Gu, Sheetal Sriram, Mathew Jacob, **Keisuke Kamahori**, Vic Li, Yi Pan, Dedong Xie, Stephanie Wang, Arvind Krishnamurthy, Baris Kasikci.
Preprint. (*SyFI Lab Blog, September 2026.*)

[P4] **Murmur: An Efficient Inference System for Long-Form ASR.**
Wei-Tzu Lee, **Keisuke Kamahori**, Baris Kasikci.
Preprint. (*Presented at MLSys 2026 YPS, May 2026.*)

[P5] **VoxServe: Streaming-Centric Serving System for Speech Language Models.**
Keisuke Kamahori, Wei-Tzu Lee, Atindra Jha, Rohan Kadekodi, Stephanie Wang, Arvind Krishnamurthy, Baris Kasikci.
Preprint.

[P6] **AgentFlux: Decoupled Fine-Tuning & Inference for On-Device Agentic Systems.**
Rohan Kadekodi*, Zhan Jin*, **Keisuke Kamahori**, Yile Gu, Sean Khatiri, Noah H. Bayindirli, Sergey Gorbunov, Baris Kasikci.
Preprint.

[P7] **Themis: Software-Defined Hardware Prefetching.**
Keisuke Kamahori, Neil Adit, Kan Zhu, Yuqi Mai, Victor Lee, Heiner Litz, Chris Kennelly, Snehasish Kumar, Hanna Alam, Milad Hashemi, David Li, Adrian Sampson, Baris Kasikci, Tipp Moseley, Parthasarathy Ranganathan, Akanksha Jain.
Preprint.

Peer-Reviewed Conferences

[C1] **M*: A Modular, Extensible, Serving System for Multimodal Models.**
Atindra Jha*, Naomi Sagan*, **Keisuke Kamahori**, Irmak Sivgin, Rohan Sanda, Steven Gao, Mark Horowitz, Luke Zettlemoyer, Olivia Hsu, Jure Leskovec, Baris Kasikci, Stephanie Wang.
NeurIPS 2026, December 2026.

[C2] **ConsumerBench: Benchmarking Generative AI Applications on End-User Devices.**
Yile Gu, Rohan Kadekodi, Sahi Chitrapu, Aradhya Agrawal, Camille Sawa, Hoang Doan Nguyen, **Keisuke Kamahori**, Yiyu Liu, Baris Kasikci.
COLM 2026, October 2026.

[C3] **TeleRAG: Efficient Retrieval-Augmented Generation Inference with Lookahead Retrieval.**
Chien-Yu Lin*, **Keisuke Kamahori***, Yiyu Liu*, Xiaoxiang Shi, Madhav Kashyap, Yile Gu, Rulin Shao, Zihao Ye, Kan Zhu, Stephanie Wang, Arvind Krishnamurthy, Rohan Kadekodi, Luis Ceze, Baris Kasikci.
MLSys 2026, May 2026.

[C4] **LiteASR: Efficient Automatic Speech Recognition with Low-Rank Approximation.**
Keisuke Kamahori, Jungo Kasai, Noriyuki Kojima, Baris Kasikci.
EMNLP 2025 (Main), November 2025. (*Featured on Hugging Face's Open ASR Leaderboard.*)

[C5] **NanoFlow: Towards Optimal Large Language Model Serving Throughput.**
Kan Zhu, Yufei Gao, Yilong Zhao, Liangyu Zhao, Gefei Zuo, Yile Gu, Dedong Xie, Tian Tang, Qinyu Xu, Zihao Ye, **Keisuke Kamahori**, Chien-Yu Lin, Ziren Wang, Stephanie Wang, Arvind Krishnamurthy, Baris Kasikci.
OSDI 2025, July 2025.

[C6] **Fiddler: CPU-GPU Orchestration for Fast Inference of Mixture-of-Experts Models.**
Keisuke Kamahori*, Tian Tang*, Yile Gu, Kan Zhu, Baris Kasikci.
ICLR 2025, April 2025. (*Also presented at PML4LRS @ ICLR 2024, May 2024.*)

[C7] **Manticore: Hardware-Accelerated RTL Simulation with Static Bulk-Synchronous Parallelism.**
Mahyar Emami*, Sahand Kashani*, **Keisuke Kamahori**, Mohammad Sepehr Pourghannad, Ritik Raj, James R. Larus.
ASPLOS 2024, April 2024.

[C8] **A 475 MHz FPGA Accelerator for RTL Simulation.**
Sahand Kashani*, Mahyar Emami*, **Keisuke Kamahori**, Mohammad Sepehr Pourghannad, Ritik Raj, James R. Larus.
FPGA 2024, March 2024.

[C9] **CiraaS: cloud computing with programmable logic.**
Kenji Tanaka, Yuki Arikawa, Tsuyoshi Ito, Yuki Matsuda, **Keisuke Kamahori**, Shinya Kaji, Takeshi Sakamoto.
SIGCOMM 2022 Poster and Demo Sessions, August 2022.

[C10] **Accelerating Decision Tree Ensemble with Guided Branch Approximation.**
Keisuke Kamahori, Shinya Takamaeda-Yamazaki.
HEART 2022, June 2022.

[C11] **GBA: Guided Branch Approximation.**
Keisuke Kamahori, Shinya Takamaeda-Yamazaki.
YArch 2022 (Co-located with ASPLOS 2022), March 2022.

Experience

University of Washington — Seattle, WA
Research Assistant, Systems for Future Intelligence (SyFI) Lab — *Jun 2023 – Present*

- Supervisor: Prof. Baris Kasikci.
- **Agents and Systems**: Developing an AI agent system for serving system development (VibeServe [P1]); an efficient fine-tuning framework for local AI agents (AgentFlux [P6]); part of the Self-Defining Systems project.
- **Systems for Multimodal Models**: Developing a universal serving system for multimodal models (M* [C1]); a streaming-centric serving system for speech LLMs (VoxServe [P5]); a compression technique for speech models (LiteASR [C4]); collaborating with AI4DeafBlind to enable fast ASR inference on low-resource accessibility devices.
- **Local AI**: Led projects on inference acceleration for RAG (TeleRAG [C3]) and MoE models (Fiddler [C6]) under resource constraints; benchmarked on-device inference (ConsumerBench [C2]).
- **Computer Architecture**: Designed software-guided optimization techniques for data center CPU prefetchers (Themis [P7]).

NVIDIA — Santa Clara, CA
Architect Intern — *Jun 2026 – Sep 2026*

- Supervisor: Joydeep Ray.
- Modeled AI inference workloads at the microarchitecture level and analyzed on-chip wiring congestion to quantify performance/design trade-offs for next-generation GPU architectures.

Kotoba Technologies Inc. — Seattle, WA / Tokyo, Japan
Founding Engineer — *Jun 2024 – Jul 2025*

- One of the earliest engineers at a startup building speech AI, with a focus on real-time speech-to-speech translation.
- Led the design and implementation of the serving infrastructure powering DOTSU, the company's core simultaneous translation product (500K+ sessions in three months), and APIs (ASR, TTS, STS), involving full-stack optimization tailored to the unique duplex architecture and strict latency requirements of real-time speech LLMs.

Fixstars Corporation — Tokyo, Japan
FPGA Engineer — *Aug 2021 – May 2023*

- Supervisor: Shinya Kaji.
- Developed an FPGA accelerator for distributed 3D Fast Fourier Transform workloads.
- Contributed to a research project on FPGA SmartNICs for data centers [C9].

The University of Tokyo — Tokyo, Japan
Undergraduate Researcher, Laboratory for Computer Architecture and Systems — *Aug 2021 – Mar 2023*

- Supervisor: Prof. Shinya Takamaeda-Yamazaki.
- Designed an acceleration method based on microarchitectural support for approximate computing [C10], [C11].

École Polytechnique Fédérale de Lausanne — Lausanne, Switzerland
Research Intern, Very Large Scale Computing Lab. — *Jul 2022 – Sep 2022*

- Supervisor: Prof. James Larus.
- Designed an FPGA accelerator for hardware simulation [C7], [C8].

Nara Institute of Science and Technology — Nara, Japan
Research Intern, Computing Architecture Lab. — *Feb 2022 – Mar 2022*

- Supervisors: Prof. Yasuhiko Nakashima and Prof. Renyuan Zhang.
- Worked on lightweight neural network processing with stochastic computing.

The University of Tokyo Tokyo, Japan
Research Intern, Matsuo Lab. *Dec 2020 – Mar 2022*

- Supervisors: Prof. Yutaka Matsuo and Ryo Okada.
- Worked on extracting human-interpretable knowledge from high-dimensional deep RL models.

Nagase Brothers Inc. Tokyo, Japan
AI Engineer *May 2020 – Aug 2021*

- Worked on developing software using deep learning techniques for educational services.

ROOT, HEP Software Foundation Remote
Google Summer of Code Student *May 2020 – Aug 2020*

- Supervisor: Dr. Oksana Shadura.
- Contributed to an open-source data analysis framework for the high-energy physics community in a Google Summer of Code project titled "Pre-conditioners applied to ROOT compression algorithms."

Awards and Grants

- People's Choice Award, Allen School Research Showcase *Oct 2026*
- MLSys 2026 Competition - NVIDIA Track (FlashInfer AI Kernel Generation Contest) *May 2026*
  1st place (GDN Track, Full-Agent), 2nd place (GDN Track, Agent-Assisted), and 3rd place (DSA Track, Full-Agent)
  Awarded NVIDIA DGX Spark, RTX 5090, and RTX 5080
- MLSys 2026 Student Travel Grant *May 2026*
- Modal for Academics Grant (USD 10,000) *Apr 2026*
- Tinker Research Grant from Thinking Machines (USD 5,000) *Feb 2026*
- Graduate Student Conference Presentation Awards from UW (USD 500) *Nov 2025*
- Allen School Travel Grant from UW CSE (USD 500) *May 2024*
- **Ph.D. Scholarship from Toyota Physical and Chemical Research Institute**
  Full tuition/insurance fee and stipend for two years (JPY 10,000,000/year) *2023 – 2025*
  Additional financial aid of USD 6,000/year *2025 – Present*
- ISCA 2022 Registration Grant from 4th uArch workshop *Jun 2022*
- ISCA 2021 Registration Grant from 3rd uArch workshop *Jun 2021*
- Travel Grant to Chulalongkorn University by JASSO *Aug 2019*
- Gold Award, Chemistry Grand Prix *2018*
- Rank A Award, Japan Mathematical Olympiad *2018*
- Rank A Award, Japan Junior Mathematical Olympiad *2016*

Mentoring

- Xikai Meng (UW ECE) *Spring 2026 – Present*
  M and VibeServe*
- Garv Ghai (Georgia Tech) *Spring 2026 – Present*
  VoxServe and multimodal serving
- Wei-Tzu (Ruby) Lee (UW ECE → Software Engineer at AWS) *Fall 2025 – Present*
  First author of Murmur (MLSys 2026 YPS)
- Tian Tang (Tsinghua University → MTS at Nex) *Spring – Summer 2024*
  Co-first author of Fiddler (ICLR 2025)

Teaching

- Guest Lecture, ECE 6950 AI Agents for Systems Engineering (Fall 2026), Cornell University
- TA, CSE 554 Machine Learning Systems (Winter 2026)

Talks

- **VibeServe: Can AI Agents Build Bespoke LLM Serving Systems?**
  At NVIDIA TRT-LLM Team, June 2026.
  At UW Mobile Intelligence Lab., May 2026.
- **Practical FPGA Development Seminar**
  At Fixstars FPGA Seminar vol. 10, May 2022.
  At Fixstars FPGA Seminar vol. 8, March 2022.
- **Accelerating Decision Tree Ensemble with Guided Branch Approximation (in Japanese).**
  At ETNET 2022, March 2022.
- **Pre-conditioners applied to ROOT compression algorithms.**
  At IRIS-HEP/GSoC Student Presentations, September 2020.

Media

- **Madrona Prize goes to UW research on agents that shrink AI models to run on small chips.**
  At GeekWire, featuring VibeSys (People's Choice Award).
- **Inside the UW Allen School: Six 'grand challenges' shaping the future of computer science.**
  At GeekWire, featuring VoxServe and ConsumerBench.

Services

Conference Reviewing

- ICLR (2027)
- AAAI (2027)
- NeurIPS (2026)
- CAIS (2026)
- ICML (2026)
- MLSys (2026)
- ICLR (2026)

Journal Reviewing

- Transactions on Machine Learning Research (TMLR)

Workshop Reviewing

- COLM 2026 Workshop on Efficient Reasoning
- CVPR 2026 Multi-Modal Reasoning for AI Agents Workshop
- ICLR 2026 MemAgents Workshop
- ICLR 2026 ES-Reasoning Workshop
- NeurIPS 2025 Workshop on Efficient Reasoning

Artifact Evaluation

- SOSP (2026)
- CAIS (2026)
- EuroSys (2026)
- MICRO (2025)

Departmental Service

- Allen School Pre-Application Mentorship Service (PAMS) Mentor *2025*
- Allen School Ph.D. Admission Committee Member *2023 – 2024*
- Allen School Ph.D. Admission Student Area Chair (Architecture) *2024*

Miscellaneous

- The Japanese Graduate Student Association in the United States (JGSAU) Committee Member *2023 – 2026*
- XPLANE SoP Writing Support Program Mentor *2023 – Present*

Skills

- **Programming Languages**: C, C++, Python, CUDA, Triton, JavaScript, Verilog, SystemVerilog, Chisel, OCaml, Scala, R
- **ML Systems & Serving**: PyTorch, vLLM, SGLang, TensorRT-LLM, FlashInfer, Hugging Face, TVM, FAISS
- **Hardware & Simulation**: Vivado, Vitis, Pynq, ChampSim, gem5

Last update: November 5, 2026